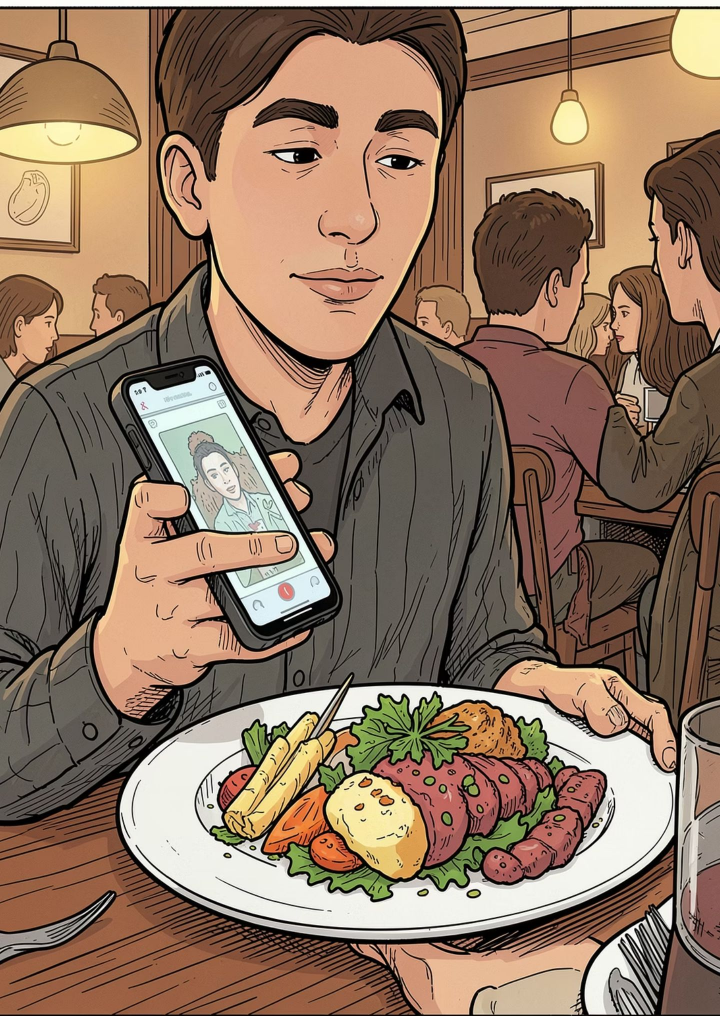


# Food Recognition & Ingredient Detection via Lightweight Neural Networks with BLIP Pseudo-Labeling

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# The Core Challenge

## The Problem

Automatic food recognition is critical for dietary tracking, but large Vision-Language Models (VLMs) like BLIP and GPT-4V are too heavy for real-time edge devices.

## The Solution

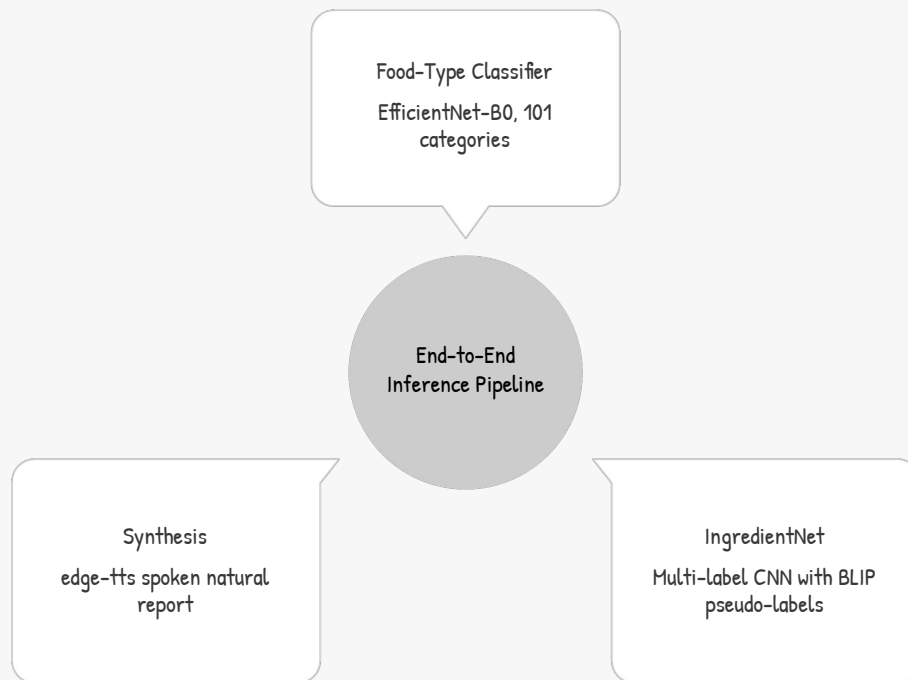
A lightweight, two-model inference pipeline achieving accurate food-type classification and multi-label ingredient detection — without any VLM overhead at runtime.

## Key Innovation

Completely eliminates VLMs during live inference by restricting them strictly to one-time offline pseudo-label generation.

# End-to-End Inference Pipeline

The system processes a single image through three parallel tracks to maximize efficiency and output richness.



All three tracks operate simultaneously on a single input image, producing a unified, spoken dietary report at the edge.

# Track 1 & Track 2: The Dual Classifier

## Track 1 – Food-Type Classifier

A fine-tuned EfficientNet-B0 identifies **101 discrete culinary categories** from a single input image with high accuracy.

## Track 2 – IngredientNet

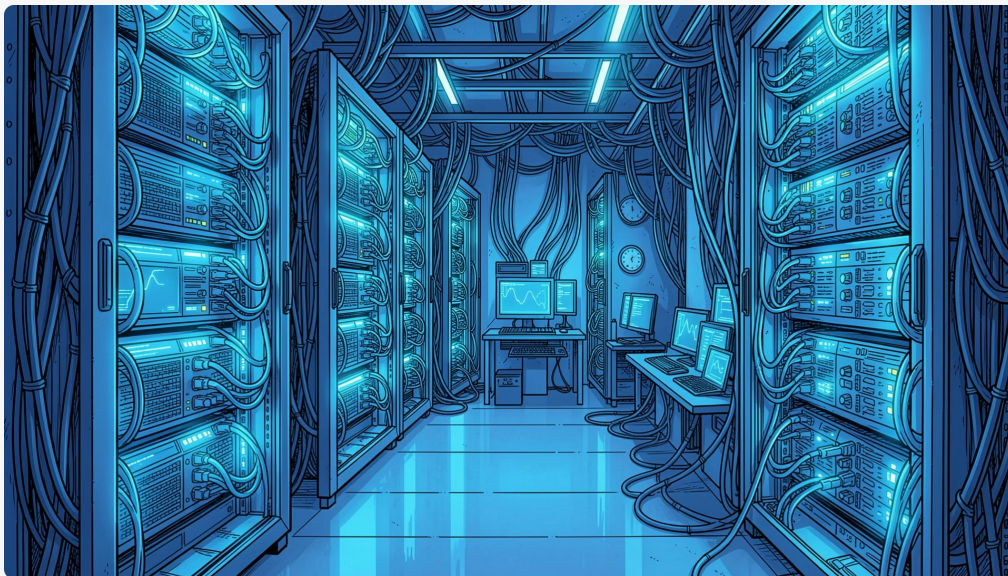
A custom multi-label CNN predicts **concurrent ingredients** simultaneously, trained entirely on BLIP-generated pseudo-labels.

## Track 3 – Synthesis

An **edge-tts engine** aggregates both outputs into a spoken natural language dietary report for accessibility.



# BLIP Pseudo-Labelling Strategy



## Offline Generation

BLIP-1 base model (~2 GB VRAM) runs offline to generate textual descriptions — avoiding the prohibitive memory costs of BLIP-2.

## Strategic Prompting

Prompts intentionally exclude the food category (e.g., *"the ingredients visible in this dish are"*), forcing visual reasoning rather than retrieval of memorized lists.

## Label Cleaning Pipeline

Removes duplicates, tokens under 3 characters, special characters, and rare ingredients (appearing fewer than 10 times). This step is critical for eliminating hallucinated tokens and repetitive phrases.

e.g. output of "beef beef beef"

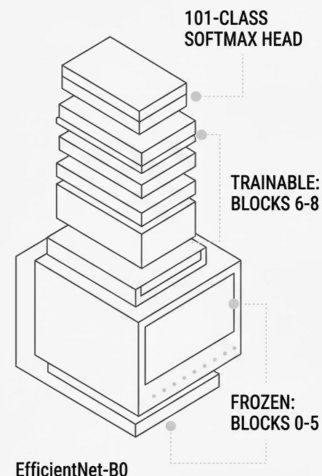
# Neural Network Architectures

## Food-Type Classifier

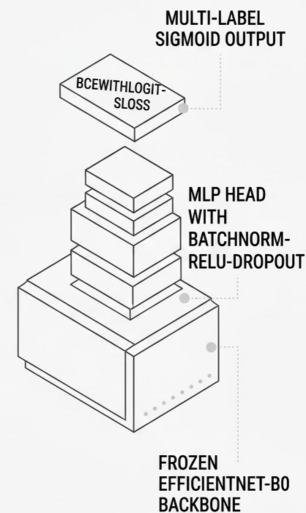
- Backbone: EfficientNet-B0, pre-trained on ImageNet-1K
- Blocks 0-5 **frozen** to preserve low-level features
- Blocks 6-8 and classification head **unfrozen** for differential fine-tuning

## IngredientNet (Multi-Label CNN)

- Fully frozen EfficientNet-B0 backbone extracting a global feature vector
- MLP head with BatchNorm, ReLU, and Dropout
- Optimized via Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss)



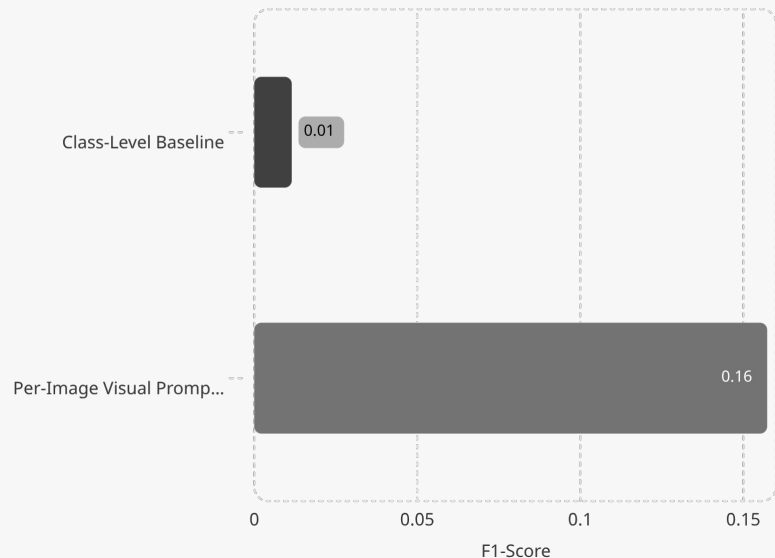
## TRACK 1: FOOD-TYPE CLASSIFIER



## TRACK 2: INGREDIENTNET

# Experimental Results: Prompting Strategy

Method



## Experiment 1 – Label Granularity

Per-image visual prompting **drastically outperforms** static class-level baselines. By generating labels specific to each image instance, the model can distinguish visually different presentations of the same dish category.

i use the test set of the blip output as the ground truth



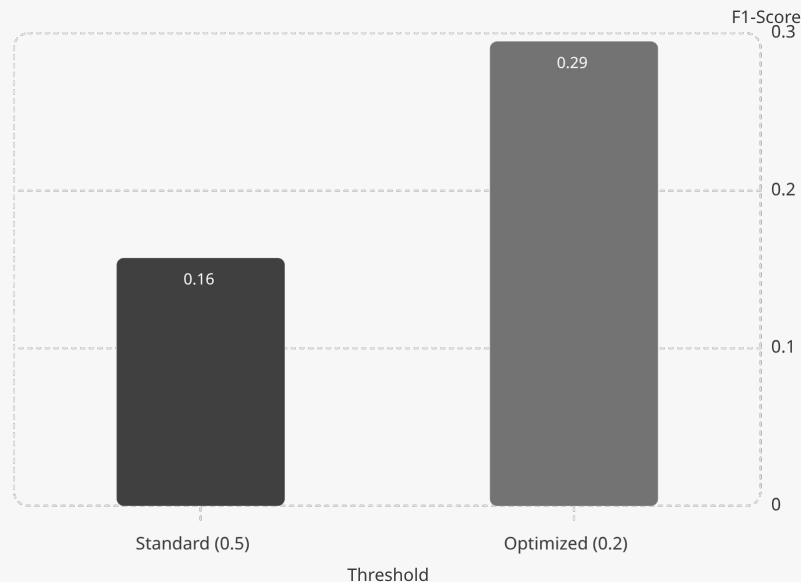
**25.5× improvement** in F1-Score — from 0.0115 to 0.2947  
— validating the per-image prompting strategy.

# Experimental Results: Threshold Sensitivity

## Experiment 2 — Decision Threshold Tuning

For large, sparse ingredient vocabularies, the standard 0.5 decision threshold is sub-optimal. Shifting the threshold post-hoc to **0.2** yielded the optimal operating point.

- Standard threshold (0.5): significantly lower F1
- Optimized threshold (0.2): **F1-Score 0.2947**
- Reflects the long-tail, sparse nature of ingredient distributions





## Key Takeaways & Edge Deployment

<100MB

Runtime Memory

Deployed system footprint — a **40×**  
**reduction** versus BLIP's ~2 GB training  
requirement.

40×

Memory Reduction

Makes real-time, on-device execution highly  
practical for consumer-grade hardware.

0

Manual Annotations

Offline VLMs plus rigorous post-processing  
completely replace manual dataset  
annotation for complex multi-label vision  
tasks.

# Summary & Contributions

01

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## Lightweight Dual-Track Pipeline

Parallel EfficientNet-B0 classifiers handle food-type and multi-label ingredient detection simultaneously.

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## Differential Fine-Tuning Strategy

Freezing early blocks preserves low-level features while adapting higher layers for domain-specific tasks.

02

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## BLIP Offline Pseudo-Labelling

Strategic prompting and rigorous label cleaning eliminate hallucinations without any manual annotation.

04

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## Edge-Ready Deployment

Under 100 MB runtime memory enables real-time dietary analysis on consumer hardware.



This work demonstrates that offline VLMs combined with rigorous post-processing can fully replace manual annotation for complex multi-label vision tasks — a blueprint for resource-constrained deployment.